

## Status — one line per numbered result

The short answer. PaperInventory.md is the long one and stays as the audit trail; this file is the thing to read first.

Measured on main after r0049, all six streams merged: build **1372 jobs, 0 errors, 0 warnings**, sorry **0**, axiom **0**, sorryAx **0**. Package: 124 modules, 46,097 lines, 2,119 theorems.

**A standing check now watches this file's class of defect.** scripts/a8-claim-check.sh asks three questions of the elaborated environment — dangling citation, proof-claim against the environment, absence-claim locus — and prints only NEW/RESOLVED against a committed baseline of **74 accepted obligations**, exiting 1 when NEW is non-empty. Cost: **12.3 s wall, 1974 MiB peak RSS**, about 5% of a cold build. It exists because r0046 measured that **seven of eight false proof-claims were true when written**: staleness produces no sorry and no build failure, so nothing else signals it. The convention for the scope-qualified sentences it checks is docs/ScopedClaims.md.

## Labels

Every paper property carries one of five labels. They answer two questions — **is it stated in Lean, and is it proved** — and the pair matters because **sorry can only see the second**.

| # | Label      | Stated?                                   | Proved?                       | Meaning                                                                                                                                                       |
|---|------------|-------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | S+P        | yes, as the paper states it               | yes                           | done                                                                                                                                                          |
| 2 | <b>S+H</b> | yes, as the paper states it               | <b>no — the proof is open</b> | a real Lean statement with a hole. H is for <i>hypothesis</i> : the result is available only conditionally, as a theorem taking the open claim as an argument |
| 3 | S≠         | yes, but <b>not the paper's statement</b> | yes                           | a weakening, an added hypothesis, or a deliberate repair of a printed defect                                                                                  |
| 4 | P          | no — prose only                           | —                             | asserted in a docstring, never under the kernel                                                                                                               |
| 5 | N          | <b>no statement at all</b>                | —                             | nothing in Lean mentions it                                                                                                                                   |

Two consequences worth stating once:

- **sorry 0 does not mean “everything is proved.”** A sorry is a hole in a proof someone *started*. Rows 4 and 5 were never started, and row 2's holes are carried as explicit hypotheses rather than as sorry, so none of the three appears in a sorry count. That is why this file reports rows 8–11 of the counts separately.
- **S≠ is not automatically a defect.** Six of the 14 are repairs of the paper's own printed errors, which is correct work; 8 are ours. r0049 re-derived the row population against the built environment rather than inheriting r0044's table, and it moved on both counts: four rows left the census (two closed in r0049, two closed in r0047 and never re-counted), and Theorem 26 moved from *ours* to *repair* once the printed statement was refuted.

## Counts

| #  |                                                             | Count                                                                                                                  |
|----|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| 1  | Numbered results in Gunter & Scott                          | <b>30, verified</b> — 16 Theorems, 13 Lemmas, 1 Proposition, 0 Corollaries; numbers 1–30 contiguous, no gap, no repeat |
| 2  | — <b>fully proved</b>                                       | <b>25</b>                                                                                                              |
| 3  | — partly proved, some part open                             | 2 (results 7, 29)                                                                                                      |
| 4  | — <b>refuted as stated</b> (our transcription or the paper) | 3 (results 26, 28, 30)                                                                                                 |
| 5  | — <b>all 30 are now stated in the tree</b>                  | 0 missing                                                                                                              |
| 6  | Paper properties (numbered conjuncts + prose claims)        | 239                                                                                                                    |
| 7  | — S+P stated and proved                                     | 169                                                                                                                    |
| 8  | — <b>S+H stated, proof open</b>                             | <b>12</b> — <i>not re-derived after r0049; carried from r0047</i>                                                      |
| 9  | — S≠ stated, but not as the paper states it                 | 14 ( <b>8 ours, 6 repairs</b> )                                                                                        |
| 10 | — P + N no Lean statement at all                            | 36                                                                                                                     |
| 11 | Prop-valued claims nothing proves                           | 7 (8 strict)                                                                                                           |

Rows 8 and 11 are the two kinds of unfinished work, and they are different: row 8 is a real Lean statement whose proof is open; row 11 is a def naming a claim **nobody attempted. sorry 0 sees neither** — which is why both are counted here and neither shows up in a build.

## The 30 numbered results

P proved · p partly proved · R refuted as stated · – not in the tree

| #  | Result                                          | Status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Theorem 1                                       | P least fixed point, and below every fixed point                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2  | <b>Theorem 2</b><br><b>(Schröder-Bernstein)</b> | P “Let $S$ and $T$ be sets. If $f : S \rightarrow T$ and $g : T \rightarrow S$ are injections, then there is a bijection $h : S \rightarrow T$ ” — printed folio 6. <b>Genuinely absent until r0048</b> ; now R48.Agent1.theorem2. The inventory had mapped it to Mathlib’s <code>Function.Embedding.schröder_bernstein</code> , but <b>that is Zermelo’s proof from Knaster–Tarski</b> , where Gunter & Scott derive it from <b>Theorem 1</b> — and <code>FixedPoint.lean</code> records that neither implies the other. The new proof follows the paper step for step, via the operator $Y \mapsto (T - f^*(S)) \cup f^*(g^*(Y))$                                                                                                                                                                                                                                                                                                                                                                                |
| 3  | Theorem 3                                       | P theorem3, theorem3_existsUnique                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 4  | Lemma 4                                         | P NormalSubposet.lean                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5  | Lemma 5                                         | P FinitaryProjection.lean                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 6  | Theorem 6                                       | P theorem6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 7  | Theorem 7                                       | p sentence 1 proved; sentences 2–3 proved only at the degenerate strength — <b>every domain has an EffectivePresentation</b> because <code>Classical.dec</code> fills its fields. The recursive forms are open: <code>Theorem7ArrowRecursive</code> , <code>Theorem7StrictRecursive</code> . r0049 restated <code>StepFunctionsDecidable</code> over <code>IsStepEnumeration</code> , quantified existentially, so the claim no longer names a guard its own hypotheses do not determine — R49.Agent3.stepFunctionsDecidable_of_compactGuard proves $\text{old} \rightarrow \text{new}$ , a <b>weakening</b> . r0049 also supplied the recursion theory the residue needed: the <code>Finset (ℕ × ℕ)</code> coding is <code>Primrec</code> , the normal-subposet search is total, and §3.2’s two conditions <b>decide boundedness and compute the join’s index</b> (R49.Agent4.computablePred_bddAbove, <code>computable_joinIdx</code> ). One item remains: <code>RecursiveNormal</code> for $K(D \rightarrow E)$ |
| 8  | Lemma 8                                         | P Product.lean                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 9  | Lemma 9                                         | P lem9_* — <b>two printed misprints repaired</b> (items 3 and 5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 10 | Lemma 10                                        | P 7 conjuncts, lem10_{prod, smash, sum, separated, lift, strict}                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 11 | Theorem 11                                      | P thm11, with thm11_converse                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 12 | Theorem 12                                      | P thm12 and the three powerdomains                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 13 | Lemma 13                                        | P lem13_smyth, lem13_hoare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 14 | Theorem 14                                      | P thm14, both directions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 15 | <b>Proposition 15</b>                           | P “A bounded complete domain is bifinite” — printed folio 31. prop15 at <code>Skeleton/Section6.lean</code> : 134, proved since that file was written. <b>It is a Proposition, not a Theorem or Lemma</b> , which is the whole reason it read as missing: both instruments that produced the gap searched only for <code>thm theorem lem lemma</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 16 | Theorem 16                                      | P thm16, thm16_positive                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 17 | Lemma 17                                        | P 10 conjuncts. r0047 <b>removed [BoundedComplete β]</b> from <code>lem17_fun</code> and <code>lem17_strictFun</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 18 | Theorem 18                                      | P thm18 — closed r0042, [ <code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code> ]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 19 | Lemma 19                                        | P lem19                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 20 | Lemma 20                                        | P lem20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 21 | Theorem 21                                      | P thm21                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 22 | Theorem 22                                      | P thm22                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 23 | Lemma 23                                        | P lem23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 24 | Lemma 24                                        | P lem24 (Gunter & Scott’s; <b>not</b> Gunter 1987’s Lemma 24, below)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 25 | Theorem 25                                      | P thm25, thm25_isUniversal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| #  | Result     | Status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 26 | Theorem 26 | R <b>refuted as printed</b><br>(R49.Agent7.not_thm26Printed_of_two_zero_aritys), and <b>proved as repaired</b> — thm26, whose added $hs : \forall i, 0 < s_i$ is therefore a <b>repair of a printed defect, not a defect of ours</b> . Theorem 26 as printed is false for any signature with <b>two or more 0-ary slots</b> , including the paper’s own worked $(2, 0, 0, 0, 0, 0)$ . The proof does not use $\text{fst}(\psi(x)) = x$ : the combinations $F_1 \dots F_n$ are quantified <i>before</i> the algebra, so one instance with $o_i = o_j$ forces $F_i = F_j$ and another with $o_i \neq o_j$ forces $F_i \neq F_j$ . It grants every printed hypothesis and asks the <i>weakest</i> conclusion — an injective homomorphism, with <code>isSubalgebraOf_range</code> proving the image is a subalgebra — so stronger readings of “isomorphic” are refuted a fortiori. The argument previously on record at <code>Combinator.lean:60–72</code> is invalid, as r0044 said: <code>isAlgEmbedding_const_of_subsingleton</code> shows two one-point algebras do land on the same subalgebra. The conclusion was right for a reason nobody had given |
| 27 | Theorem 27 | P thm27                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 28 | Lemma 28   | R <b>refuted at generic U</b> (not_forall_lemma28, witness <code>Flat Empty</code> ), and stays false after adding <code>[Domain U]</code> and <code>[BoundedComplete U]</code> . <b>Proved at U</b> : <code>lemma28AtU</code> , all nine conjuncts. The blocker is <code>UniversalForBCD U</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 29 | Theorem 29 | p sentence 1 proved (thm29). Sentence 2: <code>Thm29Second</code> <b>refuted</b> — our transcription dropped the paper’s word “domain”. <code>Thm29SecondAtDomains</code> , the true reading, is <b>open</b> . r0049 closed the finite half of what it reduces to: <b><math>\aleph</math> is universal for the finite pointed posets</b><br>(R49.Agent5.exists_normal_embedding_Ainf, new and unconditional), giving <code>thm29Normal_finiteBasis</code> . <code>Thm29Normal</code> is now open <b>only for infinite K(E)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 30 | Lemma 30   | R universal closure <b>refuted</b> (not_forall_lemma30). <code>Lemma30AtV</code> is <b>open, now at arity 1</b> — r0049 discharged <code>FpImagesBifinite V</code> outright<br>(R49.Agent6.fpImagesBifinite_V, no hypotheses, no binders), and conjuncts 1–2 now follow from <code>Thm29SecondAtDomains</code> alone. <code>Thm29Normal</code> is the single remaining named obstruction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

**Result 15 was never missing — the instruments were.** Both the declaration scan and the docstring grep searched only for `thm|theorem|lem|lemma`, and Gunter & Scott number **Propositions in the same sequence**. `scripts/numbered-status.sh` now matches `prop|proposition|cor|corollary` as well; the widening was measured before it was made, and over results 1–30 it adds exactly one hit and no false positive.

**Result 2 was genuinely absent**, and the two gaps had independent causes: 15 was a grep blind spot, 2 was a real hole. Widening the heading-word set fixes the 15-class defect but **would still have missed 2**.

Results 4, 5 and 8 carry no numbered declaration either, but are each quoted in a module docstring — `NormalSubposet.lean`, `FinitaryProjection.lean`, `Product.lean` — so **a result stated under a descriptive name is the normal case here**, and a missing number is not evidence of a missing proof.

**The figure “30 numbered results” is now verified**, by extracting every `(Theorem|Lemma|Proposition|Corollary) N` heading from the full text — ASCII headings are unaffected by the Type 3 font defect. Exactly 30, contiguous, no gap and no repeat. The single Proposition among 29 is the entire cause of this round’s measurement defect.

## What is open, in one table

| # | Item                            | What it needs                                                                                                                                                                                                                                                                                                                                                                                          |
|---|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <code>ScottHomCRecursive</code> | <b>RecursiveNormal for <math>K(D \rightarrow E)</math></b> — the one item r0049 did not close. Deciding whether two basis elements of $D \rightarrow E$ are bounded, and where their join is. <code>StepFunctionsDecidable</code> is restated and discharged from it (R49.Agent3.stepFunctionsDecidable_of_scottHomC); the order test needs no search at all (R49.Agent4.computablePred_le_stepValues) |

| # | Item                                | What it needs                                                                                                                                                                                             |
|---|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 | Theorem7ArrowRecursive              | item 1                                                                                                                                                                                                    |
| 3 | Theorem7StrictRecursive             | item 1                                                                                                                                                                                                    |
| 4 | Thm29Normal at <b>infinite K(E)</b> | the copies must nest under incl, so the type must be realized over the <i>tower's</i> image, not the $\eta$ -image. Finite bases are <b>closed</b>                                                        |
| 5 | Thm29SecondAtDomains                | item 4                                                                                                                                                                                                    |
| 6 | Lemma30AtV                          | item 4, and nothing else — FpImagesBifinite V is discharged                                                                                                                                               |
| 7 | Lem30Arrow                          | item 6                                                                                                                                                                                                    |
| 8 | PreservesRecursivePresentation      | proved at fstOp/sndOp; open at the arrow, where it is equivalent to item 2                                                                                                                                |
| 9 | the eight open $S \neq$ rows        | five of them (Universality.lem24, Universality.thm25) are <b>one</b> obstruction, not five: whether the closure image is algebraic — a closure image on an algebraic lattice is continuous, not algebraic |

**Items 2–3 and 5–7 are downstream of 1 and 4.** Only items 1, 4 and 9 are real work. Item 1 is no longer mechanical and item 4 is no longer the whole of Thm29Normal.

## Additional theories

Results from other authors, formalized here because the paper's proofs need them. None is part of Gunter & Scott's 30.

### Jung, *Cartesian Closed Categories of Domains*

| # | Result                  | Status                                                                                                                            |
|---|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 1 | Corollary 1.36          | <b>proved</b> — JungCor136, not by Jung's route (which goes through Prop. 1.22 and a retraction pair) but by indexing below cap e |
| 2 | Theorem 1.37            | <b>proved at [Domain D]</b> — R45.Agent5.thm137. Full generality open                                                             |
| 3 | Theorem 1.37 for chains | <b>proved at [Domain D]</b> — thm137Chains                                                                                        |
| 4 | Proposition 1.22        | not needed — the route around it is item 1                                                                                        |
| 5 | Theorem 2.1, 2.3        | quoted; five printed defects in Jung's write-up recorded                                                                          |

### Iwamura and Markowsky

| # | Result               | Status                                                                                                         |
|---|----------------------|----------------------------------------------------------------------------------------------------------------|
| 1 | Iwamura's lemma      | <b>proved</b> — exists_chain_directed_cover. <b>Never in Mathlib</b>                                           |
| 2 | Markowsky's theorem  | <b>proved</b> — hasChainSuprema_iff_hasDirectedSuprema. <b>Never in Mathlib</b>                                |
| 3 | HasWellOrderedInfima | <b>no producer exists</b> — five occurrences, all consumers. A complete reduction chain with an empty left end |

### Spreen 2005

| # | Result    | Status                                                                                                   |
|---|-----------|----------------------------------------------------------------------------------------------------------|
| 1 | Lemma 5.8 | <b>proved</b> — PropertyM.hasOmegaOpBoundsAbove_pair. This, not Iwamura, is what closed Jung's 1.37 here |

### Gunter 1987, *Universal Profinite Domains*

| # | Result           | Status                                                                                                                                                                                                        |
|---|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Lemma 24 at M(A) | <b>proved</b> — R47.Agent1.lemma24_MPair. <b>The first proof anywhere:</b> Gunter's printed Lemma 24 produces <i>some</i> $A^+$ and is not about M(A); the M(A) form is a remark with no theorem and no proof |

| # | Result                             | Status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 | HasNormalRealizations at Ainf      | <b>refuted</b> — not_hasNormalRealizations_Ainf. The route to Thm29Normal is sound but its hypothesis is unsatisfiable at this tower. <b>r0049 narrowed what this closes</b> : the refutation quantifies <i>universally</i> over finite normal $A \triangleleft A^\infty$ , while Thm29Normal quantifies <i>existentially</i> over embeddings, so it closes <b>extension</b> of an embedding already fixed in $A^\infty$ and not <b>construction</b> of one. That is what let lemma24_Step be run inside the stages |
| 3 | Proposition 21, Theorems 22 and 25 | used, via the implication thm29Normal_of_hasNormalRealizations                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## Defects in the printed paper

**Ten**, nine recorded in StatementRecovery.md. The tenth is r0049's: **Theorem 26 is false as printed** for any signature with two or more 0-ary slots, kernel-checked by R49.Agent7.not\_thm26Printed\_of\_two\_zero\_aritys. That row sat in the suspected-and-refuted table below until r0049 because **the reason on record for it was wrong**, which is a separate defect from the claim being wrong.

Two suspected defects remain **our** transcription errors, not the paper's:

| # | Suspected                          | Actually                                                                                                                                                                     |
|---|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Theorem 29's second sentence false | our transcription dropped "domain" from "any bifinite domain"                                                                                                                |
| 2 | the step-function enumeration      | our guard tests compactness where the paper tests boundedness — and r0049 sharpened this: on the paper's own index the join always exists, so <b>the paper tests neither</b> |

An **eleventh candidate is not yet recorded**: r0049's agent3 reports that the same folio-12 sentence writes  $\bigcup\{f(y) \mid y \in \mathbb{N} \rightarrow x\}$  where  $f$  is bound nowhere and  $s$  is meant. It is not counted above and not yet in StatementRecovery.md.

## Prose defects, and what r0049 measured about them

The sweep of documentation claims reached 264 sites. The useful result is a **negative** one: 183 was the wrong denominator. 85 are statements about the proof script directly below them, already checked by the fact that it elaborates; 7 are Gunter & Scott's own sentences; 114 are scope-qualified (ScopedClaims.md); **only 43 are decidable against the environment**, and those are now decided in bulk every round rather than by hand.

Of the 18 adjudicated, **9 came out false**. Among the absence claims — sentences asserting the package lacks something — the enrichment is sharp: **7 of the 8 package-scoped rows are false, 87.5% against r0046's 20.8% base rate**. The best specimen is PropertyM.lean:845, refuted by a declaration 100 lines below it *in its own file*, missed by r0046's intra-file detector for one reason — its subject is a noun phrase rather than a backticked name. **0 of 55 absence claims named a scope before r0049**.

Two open defects in the instruments themselves, both measured rather than suspected:

| # | Instrument            | Defect                                                                                                                                                                                                                                                                                         |
|---|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | a8-claim-check.sh     | the baseline keys obligations on file:line, so an insertion above a standing obligation reports it as NEW + RESOLVED. First post-merge run: 1 of each, and they are the <b>same</b> claim, moved from FunctionSpace.lean:527 to :617 by a docstring edit. Key on normalized claim text instead |
| 2 | a5-r47-conditional.sh | hardcoded R00T to the agent5 worktree, so every run after r0047 measured that checkout rather than the caller's. Fixed in r0049 to resolve from the script's own location; <b>row 8 has still not been re-derived since r0047</b>                                                              |

## Reproducing

```
scripts/counts.sh          # modules, lines, theorems, sorry
scripts/numbered-status.sh # declarations per numbered result
scripts/a6-env-scan.sh <out> # then a6-summarize.py for row 11
scripts/a5-r47-conditional.sh # row 8, the S+H conditional surface
```

```
scripts/a8-claim-check.sh          # standing prose-staleness check
scripts/compile.sh -r <round>      # build
```

counts.sh, numbered-status.sh and a5-r47-conditional.sh are bash; axioms.sh, a6-env-scan.sh and md2pdf.sh are **zsh** and fail under bash — `${0:A:h}` and unquoted-array word-splitting are zsh-only.